Ria Banerjee

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SUMMARY: Backend Software Engineer with 4 years building production AI-powered systems and distributed microservices. Specialized in event-driven architectures, LLM applications (RAG, AI agents, MCP server), and cloud infrastructure on AWS/Kubernetes.

TECHNICAL SKILLS

- *Languages:* Python, Java, SQL, Go, C++.
- *AI/ML & LLM:* LangChain, LangGraph, CrewAI (multi-agent), RAG (Retrieval-Augmented Generation), Prompt Engineering, Fine-tuning, Vector DBs (pgvector), vLLM, MCP servers, OpenAI API, Claude code, AWS Bedrock.
- *Backend and API: FastAPI, Django REST, Flask, SpringBoot, Git, gRPC. REST.*
- *Web Development:* TypeScript, React, JavaScript, Node.
- *Messaging & Streaming:* Kafka, RabbitMQ, Redis Streams.
- *Cloud, Infrastructure, CI/CD:* AWS (EKS, ECS, S3, ALB), Kubernetes, Docker, Terraform, Helm, Ansible, Jenkins.
- *Database*: MongoDB, PostgreSQL, MySQL, Vector database (Supabase).

WORK EXPERIENCE

SWE Backend & Infrastructure, EAGL Technology Inc. *May 2024 – Present* | *Albuquerque, USA*

Gunshot detection platform deployed serving Fortune 500 companies and areas of public safety (schools, hospitals).

- Scaled real-time gunshot detection platform to **90K+ IoT sensors** using event-driven microservices (**Python**, **Django REST, Kafka, PostgreSQL)** achieving under 250ms p99 latency.
- Designed and prototyped RAG-based support chatbot using OpenAI model integrated with system documentation, enabling real-time query responses for system documentation and operational data.
- Created internal database-querying application using a crew of **AI agents and MCP servers** (PostgreSQL, Fileserver) to reduce lookup times and faster development of the team, hosted with Mistral on Lambda Cloud.
- Built multi-tenant WebSocket infrastructure handling 8K+ concurrent connections across 15 enterprise deployments with **Redis** Pub/Sub routing and 99.9% delivery on **AWS EKS.**
- Architected **MySQL** binlog-based CDC pipeline for air-gapped infrastructure achieving real-time replication, eliminating 4+ hour batch delays and enabling sub-second consistency.
- Deployed AWS infrastructure (EKS, ALB, S3, Route 53) with Terraform IaC enabling zero-downtime blue-green deployments and distributed rate limiting (Redis), maintaining 99.995% uptime.
- Reduced MTTR from 45 to 8 minutes by implementing observability stack (**Prometheus, Grafana, Jaeger**) with distributed tracing, identifying 25+ critical bottlenecks.
- Deployed production **RabbitMQ+MQTT** cluster on **Kubernetes** handling 100K+ concurrent IoT connections with automatic failover and message persistence.

Founding Software Engineer, BabelCast Inc. *October 2023 – January 2024* | *Remote, USA*

Real-time voice augmentation software.

- Scaled voice AI platform from 0 to 5K+ concurrent users in 3 months by architecting Django REST microservices with API Gateway, horizontal autoscaling on ECS, and request throttling handling 10K+ API calls daily.
- Implemented enterprise-grade authentication with JWT + OAuth2 (Okta), role-based access control, and SSO integration securing the platform.
- Reduced voice model query latency 25% (320ms → 240ms) by designing GraphQL API with DataLoader batching, Redis caching layer, and optimized database queries serving 5K+ voice models.

System Engineer, Equifax (Tata Consultancy Services) *May 2021 – August 2022* | *India*

- Contributed to the development of Spring Boot microservices handling large-scale credit data workflows.
- Worked with Apache Kafka to process streaming data and ensure reliable message delivery between services.
- Supported deployment and monitoring of Java services in Kubernetes-based environments.
- Participated in code reviews and followed enterprise coding and security standards.

EDUCATION

University of North Carolina Charlotte, USA | Masters in Computer Science *August 2022-May 2024*

Nirma University, India | Masters of Computer Applications *August 2019-May 2021*

Ahmedabad University, India | Bachelors of Computer Applications *August 2016-May 2019*

RESEARCH & PROJECTS

Graduate Research Assistant, University of North Carolina, Charlotte *February 2023-October 2023*

- Designed Regression model using Tensorflow to analyze real-time bio-medical data collected with Meta Quest.

Logs analysis platform

- Agentic AI platform using CrewAI and LangGraph to analyze log files and chat in realtime about the logs.

Personal Portfolio website (riabanerjee.dev)

- Built a web portfolio using React, Python, and OpenAI API, hosted on AWS EC2.